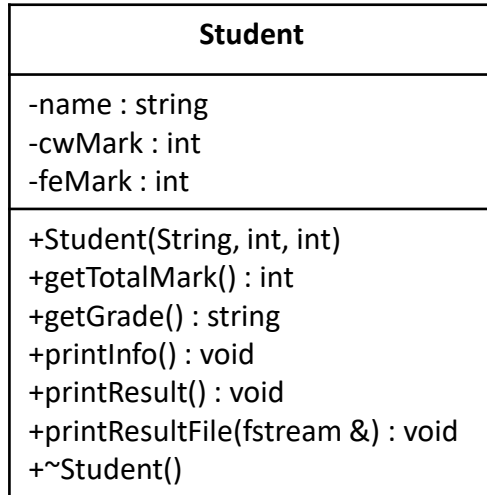


SECJ2013 - DSA

2024/2025-1

LAB 1

Write C++ header file (Student.h) to implement the Student class as presented by the below class diagram:



- ❑ The `getTotalMark` function return the total of coursework and final exam marks (`cwMark + feMark`)
- ❑ The `getGrade` function return the grade based on total mark returned by the `getTotalMark` function.
Below is the mark ranges and their corresponding grades:
75 – 100 => A
65 – 74 => B
50 – 64 => C
35 – 49 => D
0 – 34 => E
- ❑ Example of screen output for `printInfo` function:

```
Name: Adam  
Coursework: 60  
Final Exam: 20
```
- ❑ Example of screen output for `printResult` function (name, total mark, and grade):

```
Adam 80 A
```
- ❑ The `printResultFile` function should produce an output as `printResult` function but in the form of text file
- ❑ The example of screen output of the destructor

```
Destroy student object - Adam
```

Write a C++ program (Lab1.cpp) that will do the followings:

- ❑ Declare an array to store the list of pointer of Student object:

```
const int LIST_SIZE = 10;
Student* studList[LIST_SIZE];
```
- ❑ Read the input of student name, coursework mark, and final exam mark from the text file ("Marks.txt")
- ❑ Create and set the parameters of Student object by using the input data read from the "Marks.txt" file. The Student object must be stored as a pointer inside the `studList` array
- ❑ Iterate `studList` array to display the student information (name, coursework mark, final exam mark)
- ❑ Iterate `studList` array to print out the student final results (name, total mark, grade) on the screen and into the file ("Results.txt")

Results.txt

```
Adam 80 A
Brady 47 D
Cindy 60 C
```

Marks.txt

```
Adam 60 20
Brady 30 17
Cindy 38 22
```

Screen output of Lab1.cpp:

```
Student mark info:
Name: Adam
Coursework: 60
Final Exam: 20

Name: Brady
Coursework: 30
Final Exam: 17

Name: Cindy
Coursework: 38
Final Exam: 22

Print and save results to file:
Adam 80 A
Destroy student object - Adam
Brady 47 D
Destroy student object - Brady
Cindy 60 C
Destroy student object - Cindy
```